Week 2, Lecture 1
Photon statistics

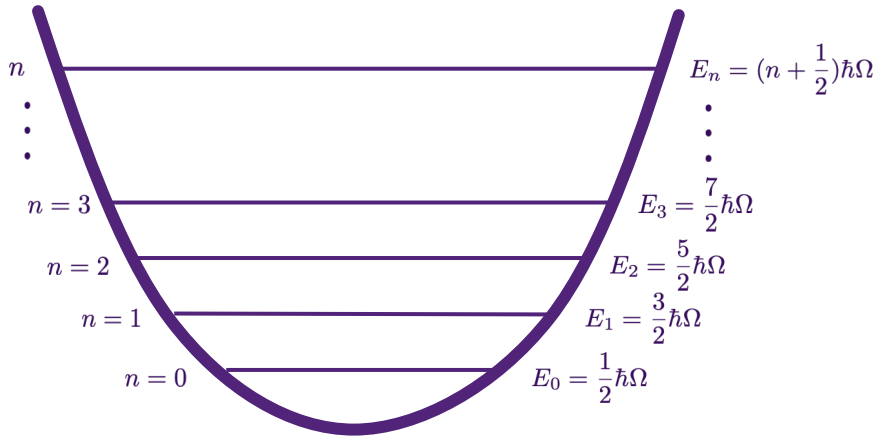
Lecturer: Josephine
Email: j.dias@uq.edu.au
Office: Parnell Room 320
Office hours: Thursdays 3-4pm

- (Photon statistics) Sec. 4.3.3, page 108 of Bachor, Hans-A. / Ralph, Timothy C. A guide to experiments in quantum optics (2019)
- (Photon statistics) Sec 5.1-5.6 of Fox, M. Quantum optics – an introduction (2006)
- (Thermal states) Sec. 3.2.3 page 45 of Leonhardt, U. Essential Quantum Optics (2010)

Review

What you learned last week…

Quantized single mode EM field → **Simple harmonic oscillator**



Hamiltonian: $\hat{H}=\hbar\Omega\left(\hat{a}^\dagger\hat{a}+\frac{1}{2}\right)$

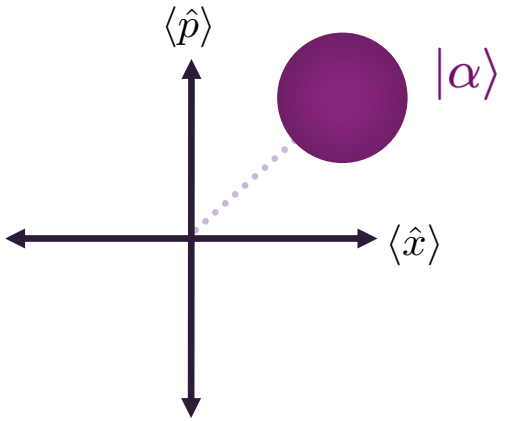
Eigenstates: $\{|n\rangle\}=\{|0\rangle,|1\rangle,|2\rangle,\cdots\}$

Annihilation and creation operators

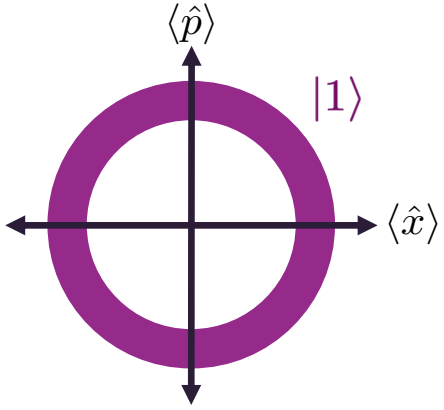
$$\hat{a}\,|n\rangle=\sqrt{n}\,|n-1\rangle \qquad [\hat{a},\hat{a}^\dagger]=1$$
$$\hat{a}^\dagger\,|n\rangle=\sqrt{n+1}\,|n+1\rangle$$

Phase space representation:

$$\hat{x}=\hat{a}+\hat{a}^\dagger$$
$$\hat{p}=i(\hat{a}^\dagger-\hat{a})$$



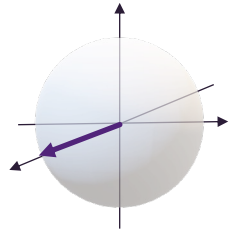
Coherent state



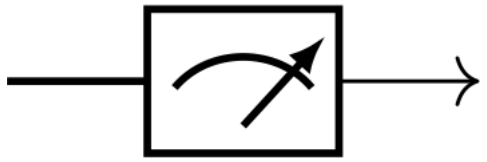
Number state

Ensembles of quantum states

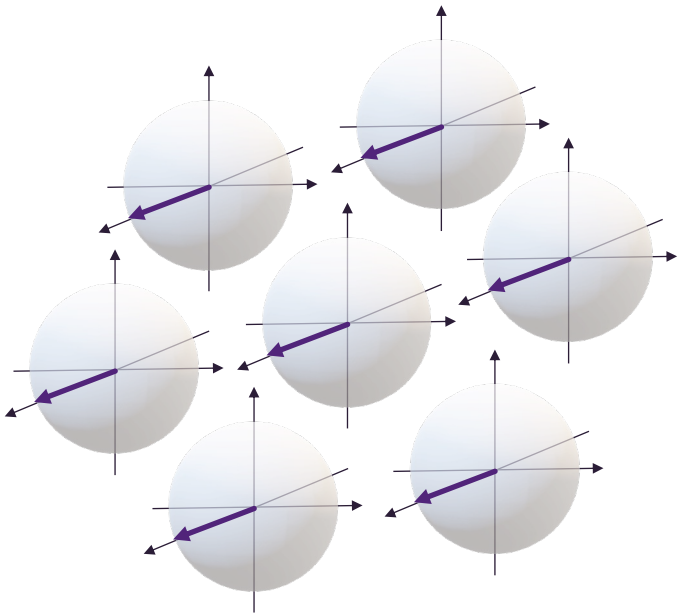
$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)$$

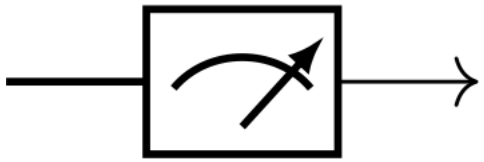


measure in basis
$\{|0\rangle, |1\rangle\}$



Either $|0\rangle$
or $|1\rangle$

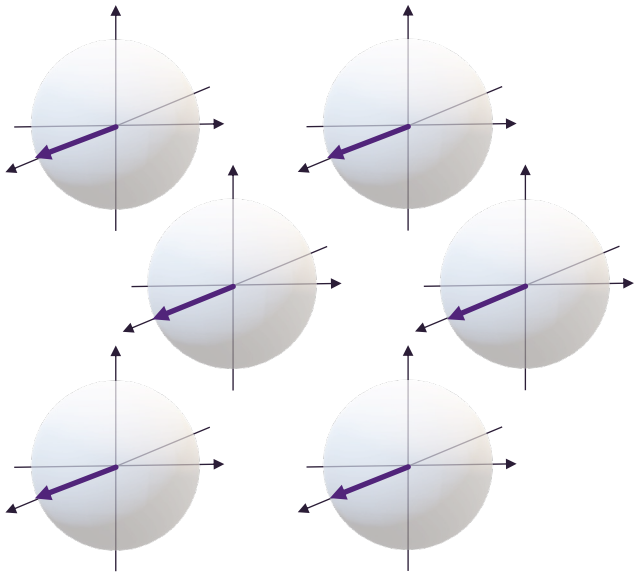




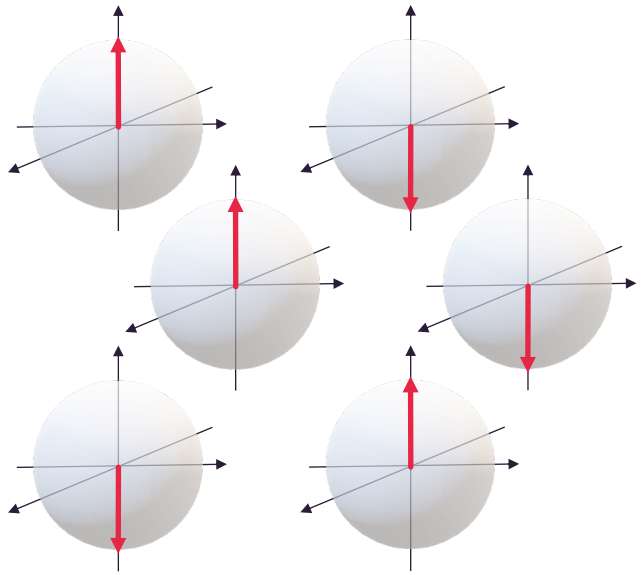
50% $|0\rangle$
50% $|1\rangle$

Pure vs. mixed ensembles

Pure

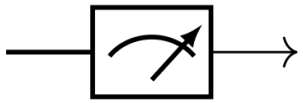


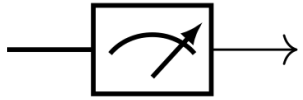
Mixed



How can we describe this ensemble?

measure in basis $\{|0\rangle, |1\rangle\}$





measurement results	50% $\|0\rangle$ and 50% $\|1\rangle$	50% $\|0\rangle$ and 50% $\|1\rangle$
ensemble	all $\|+\rangle$	50% $\|0\rangle$ and 50% $\|1\rangle$

Density operator

To describe mixed ensembles of quantum states, we use density operators defined as:

$$\hat{\rho} = \sum_i p_i \, |\psi_i\rangle \, \langle\psi_i|$$

p_i: probability of being in state $|\psi_i\rangle$

For a pure state:

$$\hat{\rho} = |\psi\rangle \, \langle\psi|$$ ← only one state ket

Dimension of ρ (for a single qubit):

$$\hat{\rho} = \sum_i p_i \, |\psi_i\rangle \, \langle\psi_i|$$

column vector: 2×1

row vector: 1×2

$\Rightarrow \rho$ is a 2×2 matrix

- p_i are classical probabilities, and they must sum to one

$$\sum_i p_i = 1$$

- $|\psi_i\rangle$'s don't need to be orthogonal
- A *qubit* is a two-level quantum system
- A pure state of a single qubit can be represented by a 2×1 column vector
- A mixed state of a single qubit can be represented by a 2×2 square matrix

Density operator

1. ρ **is Hermitian**
2. $\mathrm{Tr}[\rho] = 1$
3. ρ **is positive semidefinite**
4. $\mathrm{Tr}[\rho^2] \leq 1$ **(if** ρ **is pure then** $\mathrm{Tr}[\rho^2] = 1$ **)**

- All eigenvalues of a Hermitian square matrix are real
- Positive semidefinite: all eigenvalues are non-negative
- The purity of a quantum state is $Tr[\hat{\rho}^2]$

  For pure states: $Tr[\hat{\rho}^2]$=1

Evolution of the density operator

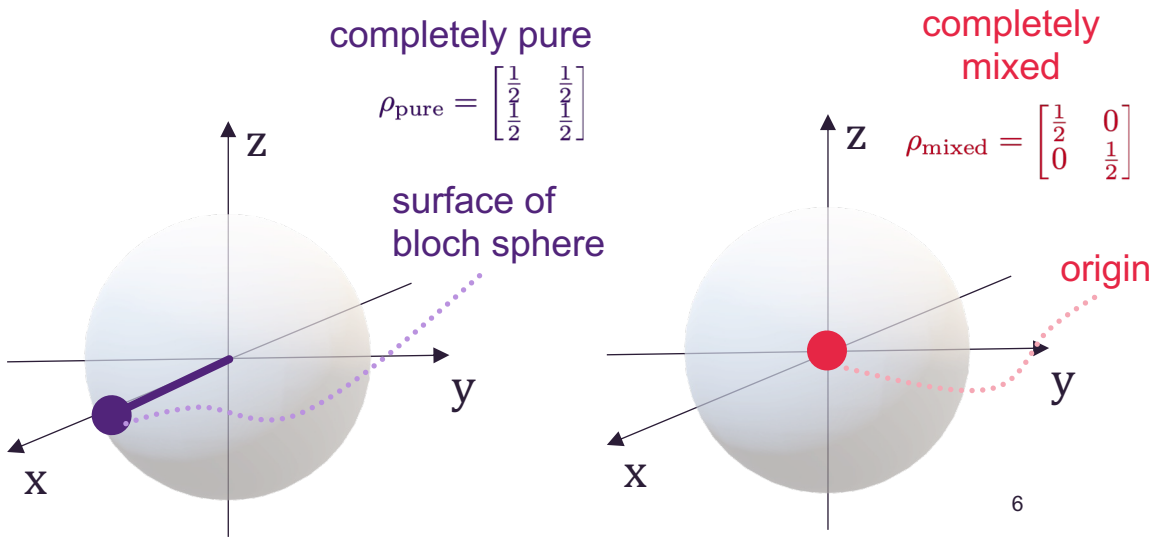
$$|\psi_{\mathrm{out}}\rangle = \hat{U}\,|\psi_{\mathrm{in}}\rangle \quad \text{pure}$$

$$\hat{\rho}_{\mathrm{out}} = \hat{U}\,\hat{\rho}_{\mathrm{in}}\hat{U}^{\dagger} \quad \text{mixed}$$

Expectation values

$$\langle O\rangle = \langle\psi|\,O\,|\psi\rangle \quad \text{pure}$$

$$\langle O\rangle = \mathrm{Tr}\,[O\rho] \quad \text{mixed}$$



Entropy of a quantum system

- **Classical entropy**

$$S = -k_B \sum_i p_i \ln p_i$$

- **Quantum**

$$S = -k_B \mathrm{Tr}\left[\hat{\rho} \ln \hat{\rho}\right]$$

- **Spectral decomposition**

$$\hat{\rho} = \sum_i \lambda_i \left|\phi_i\right\rangle \left\langle\phi_i\right|$$

eigenvalues

eigenvectors



$$S = -k_B \mathrm{Tr}\left[\hat{\rho} \ln \hat{\rho}\right] = -k_B \sum_i \lambda_i \ln \lambda_i$$

Thermal states

- **EM radiation emitted by a hot body is called *thermal light* or black-body radiation**
- **Eg.**
  - Sunlight
  - Resistive filaments
- **Here, we again focus on a single mode (monochromatic light)**

- **Thermal states maximise entropy S for a given energy E**

$$\hat{\rho} = \sum_n p_n \left|n\right\rangle \left\langle n\right|$$

- **Maximise entropy S in the presence of constraints using Lagrange multipliers**

$$S = -k_B \underbrace{\sum_n p_n \ln p_n}_{\text{entropy}} - a \underbrace{\left(\sum_n p_n - 1\right)}_{\text{normalisation}} - b \underbrace{\left(\sum_n p_n E_n - E\right)}_{\text{energy}}$$

See page 45 of Leonhardt, U. Essential Quantum Optics (2010) for derivation

Thermal states

$$\rho = \frac{1}{Z}\sum_{n=0}^{\infty} e^{-\hbar\Omega n/kT}\,|n\rangle\langle n| \qquad Z = \sum_{n=0}^{\infty} e^{-\hbar\Omega n/kT} = \frac{1}{1-e^{-\hbar\Omega/kT}}$$

$$\rho = \left(1-e^{-\hbar\Omega/kT}\right)\sum_{n=0}^{\infty} e^{-\hbar\Omega n/kT}\,|n\rangle\langle n|$$

Boltzmann distribution

Statistical mixture of Fock states

- **Probability of n photons** $p_n = \left(1-e^{-\hbar\Omega/kT}\right)e^{-\hbar\Omega n/kT}$

Thermal states

- **Expectation value of $\hat{n}$**

$$\rho = \left(1 - e^{-\hbar\Omega/kT}\right) \sum_{n=0}^{\infty} e^{-\hbar\Omega n/kT} \left|n\right\rangle \left\langle n\right|$$

$$p_n = \left(1 - e^{-\hbar\Omega/kT}\right) e^{-\hbar\Omega n/kT}$$

$$\bar{n} = \sum_{n=0}^{\infty} p_n n = \sum_{n=0}^{\infty} n x^n (1-x) = \frac{x}{1-x} \qquad x = e^{-\hbar\Omega/kT}$$

$$\bar{n} = \frac{1}{x^{-1} - 1}$$

$$\bar{n} = \frac{1}{e^{\hbar\Omega/kT} - 1}$$

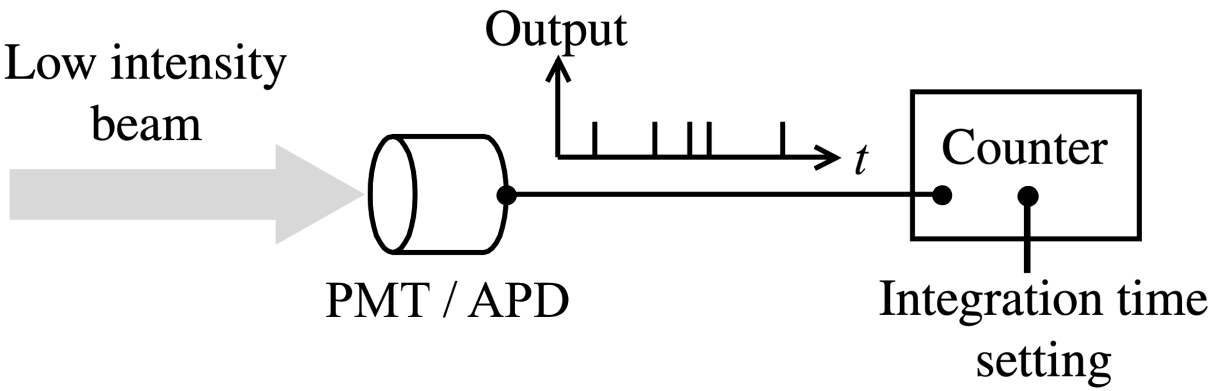
$$\hat{\rho} = \frac{1}{\bar{n}+1} \sum_{n} \left(\frac{\bar{n}}{\bar{n}+1}\right)^n \left|n\right\rangle \left\langle n\right|$$

Photon statistics

We will talk more about photodetection in lecture 3 this week...

For now, the observable we are measuring is

$$\hat{n} = \hat{a}^\dagger \hat{a}$$



- **Probability of detecting n photons**

$$p_n = |\langle n|\psi\rangle|^2$$

- **Average number of photons**

$$\bar{n} = \langle \hat{n} \rangle = \langle \psi | \hat{a}^\dagger \hat{a} | \psi \rangle$$

- **Variance**

$$\mathrm{var}\,(n) = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2$$

Photon statistics

Coherent light – photon number distribution

- **Coherent state expansion in number basis**

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

- **Probability of detecting n photons**

$$p_n = |\langle n|\alpha\rangle|^2 = \langle\alpha|n\rangle \langle n|\alpha\rangle$$

$$\langle n|\alpha\rangle = e^{-|\alpha|^2/2} \frac{\alpha^n}{\sqrt{n!}}$$

$$p_n = |\langle n|\alpha\rangle|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}$$

$$p_n = e^{-\bar{n}} \frac{\bar{n}^n}{n!}$$

Poissonian distribution

- **Expectation value of $\hat{n}$**

$$\bar{n} = \langle\alpha|\hat{n}|\alpha\rangle$$

$$\hat{n} = \hat{a}^\dagger \hat{a}$$

$$\bar{n} = \langle\alpha|\hat{a}^\dagger \hat{a}|\alpha\rangle$$

$$\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle$$

$$\langle\alpha|\,\hat{a}^\dagger = \alpha^*\,\langle\alpha|$$

$$\bar{n} = |\alpha|^2$$

Photon statistics

Coherent light – photon number variance

- **Photon number variance**

$$\mathrm{var}\,(n) = \langle \Delta \hat{n}^2 \rangle = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2$$

$$\langle \hat{n}^2 \rangle = \langle \alpha | \hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a} | \alpha \rangle$$

$$\langle \hat{n}^2 \rangle = |\alpha|^2 \langle \alpha | \hat{a} \hat{a}^\dagger | \alpha \rangle$$

$$\hat{a} |\alpha\rangle = \alpha |\alpha\rangle$$

$$\langle \alpha | \hat{a}^\dagger = \alpha^* \langle \alpha |$$

$$[\hat{a}, \hat{a}^\dagger] = 1 \qquad \hat{a}\hat{a}^\dagger = 1 + \hat{a}^\dagger \hat{a}$$

$$\langle \hat{n}^2 \rangle = |\alpha|^2 \langle \alpha | (1 + \hat{a}^\dagger \hat{a}) | \alpha \rangle$$

$$\langle \hat{n}^2 \rangle = |\alpha|^2 (1 + |\alpha|^2)$$

$$\langle \hat{n}^2 \rangle = |\alpha|^4 + |\alpha|^2$$

$$\mathrm{var}\,(n) = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2$$

$$\mathrm{var}\,(n) = |\alpha|^4 + |\alpha|^2 - (|\alpha|^2)^2$$

$$\mathrm{var}\,(n) = |\alpha|^2 = \bar{n}$$

- **Variance is equal to mean**

Poissonian distribution

Photon statistics

Coherent light - summary

- **Coherent state expansion in number basis**

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

- **Probability of detecting n photons**

$$p_n = |\langle n|\alpha\rangle|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}$$

- **Average number of photons in coherent state**

$$\bar{n} = \langle\alpha|\hat{a}^\dagger \hat{a}|\alpha\rangle = |\alpha|^2$$

- **Photon number distribution**

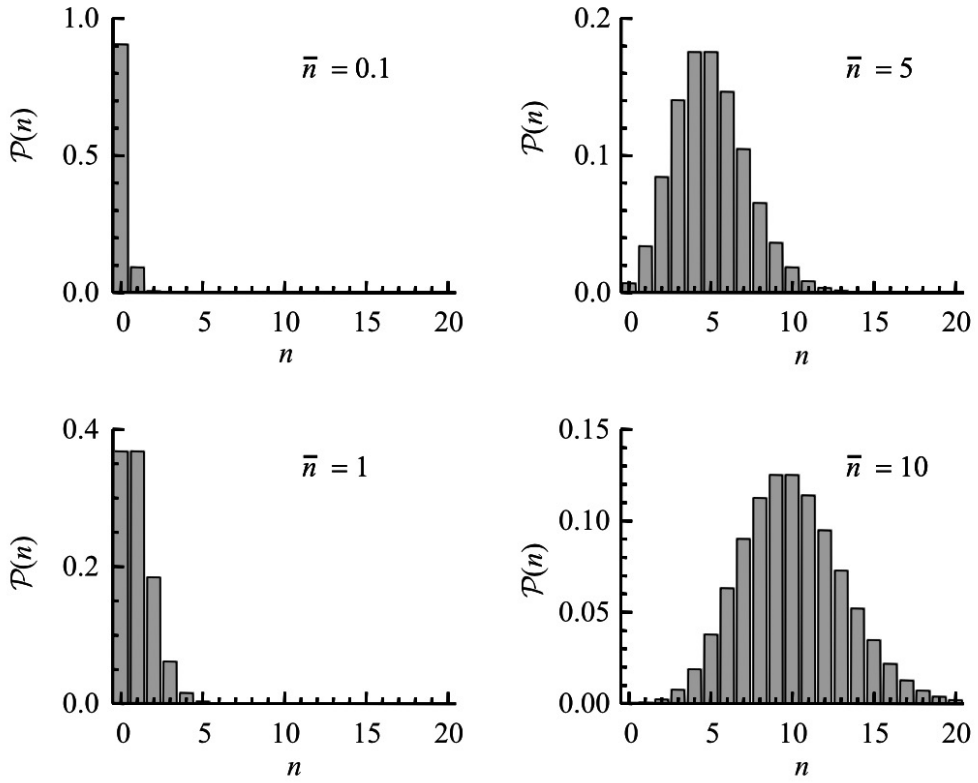
$$p_n = e^{-\bar{n}} \frac{\bar{n}^2}{n!}$$

Poissonian distribution

- **Variance is equal to mean**

$$\mathrm{var}\,(n) = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2 = \bar{n}$$

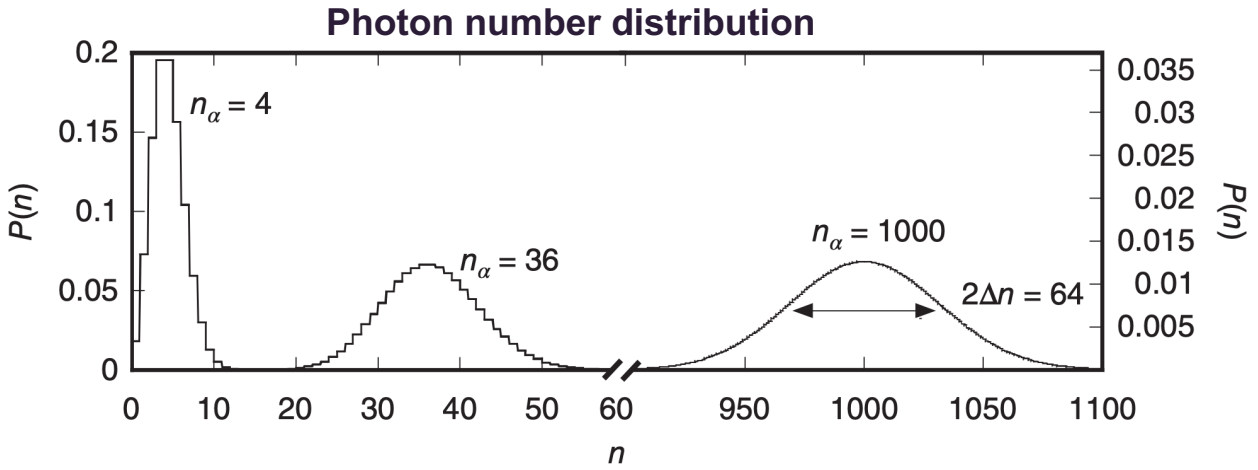
Photon number distribution



p. 81 Fox, M. Quantum optics – an introduction (2006)

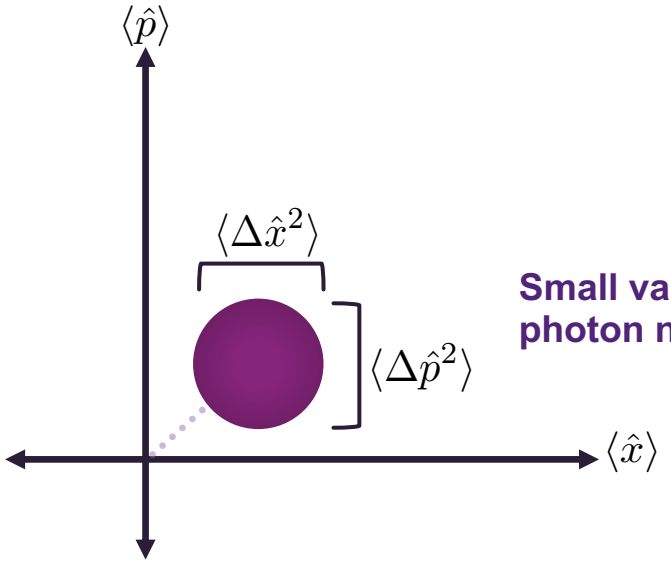
Photon statistics

Coherent light



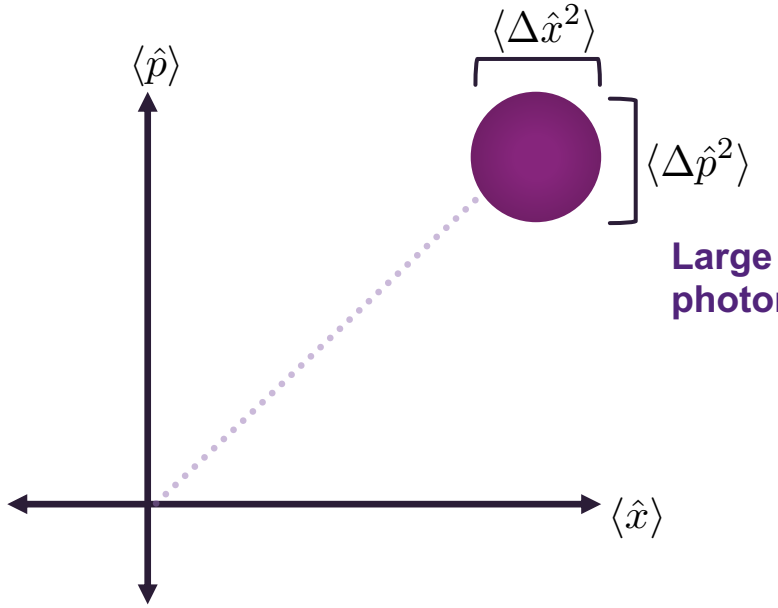
p. 109 Bachor, Hans-A. / Ralph, Timothy C.
A guide to experiments in quantum optics (2019)

- **Coherent states in phase space**



$$\langle\Delta\hat{x}^2\rangle = \langle\Delta\hat{p}^2\rangle = 1$$

Small variance in photon number



Large variance in photon number

Photon statistics

Thermal states

$$\rho = \left(1 - e^{-\hbar\Omega/kT}\right) \sum_{n=0}^{\infty} e^{-\hbar\Omega n/kT} \left|n\right\rangle \left\langle n\right|$$

- **Variance**

$$\mathrm{var}\,(n) = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2$$

$$\langle \hat{n}^2 \rangle = \sum_{n=0}^{\infty} p_n n^2$$

$$= \sum_{n=0}^{\infty} n^2 x^n (1-x) = \frac{x^2 + x}{(x-1)^2}$$

$$p_n = \left(1 - e^{-\hbar\Omega/kT}\right) e^{-\hbar\Omega n/kT}$$

$$x = e^{-\hbar\Omega/kT}$$

$$\bar{n} = \langle \hat{n} \rangle = \frac{x}{1-x}$$

$$\mathrm{var}\,(n) = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2 = \frac{x^2 + x}{(x-1)^2} - \left(\frac{x}{1-x}\right)^2$$

$$\mathrm{var}\,(n) = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2 = \bar{n}^2 + \bar{n}$$

Photon statistics

Thermal states

- **Probability of n photons**

$$p_n = \left(1 - e^{-\hbar\Omega/kT}\right) e^{-\hbar\Omega n/kT}$$

- **Average photon number**

$$\langle \hat{n} \rangle = \sum_{n=0}^{\infty} p_n n \qquad \bar{n} = \frac{1}{e^{\hbar\omega/k_B T} - 1}$$

- **Variance**

$$\langle \hat{n}^2 \rangle = \sum_{n=0}^{\infty} p_n n^2$$

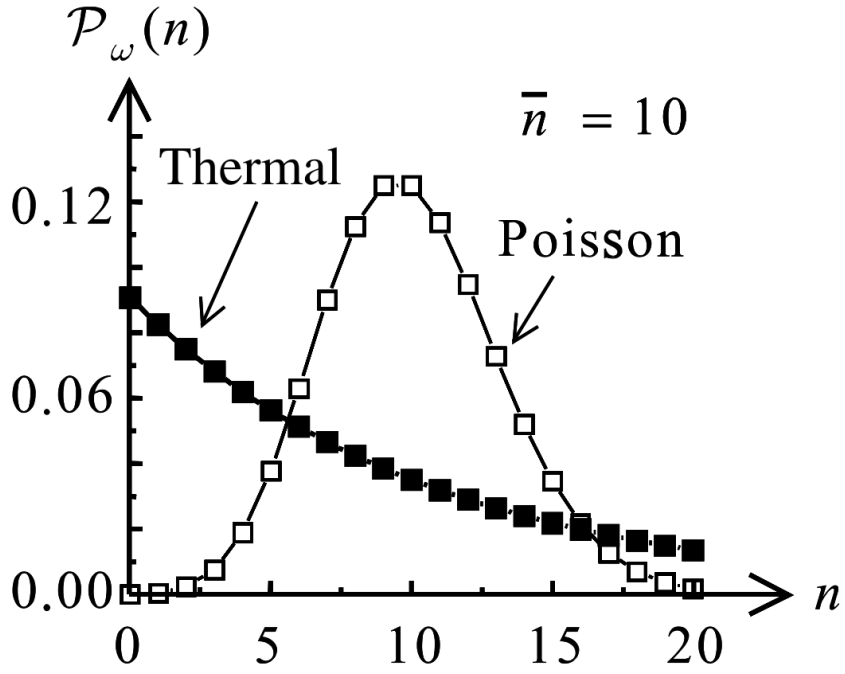
$$\mathrm{var}\,(n) = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2 = \bar{n}^2 + \bar{n}$$

- **Variance is larger than mean**

Super-poissonian statistics

Super-poissonian statistics: can be explained by a semi-classical model

Photon number distribution



p. 85 Fox, M. Quantum optics – an introduction (2006)

$|0\rangle$ **is the most probable state for thermal light**

Photon statistics

Photon number states

- **Average photon number**

$$\bar{n} = \langle n|\hat{n}|n\rangle = n$$

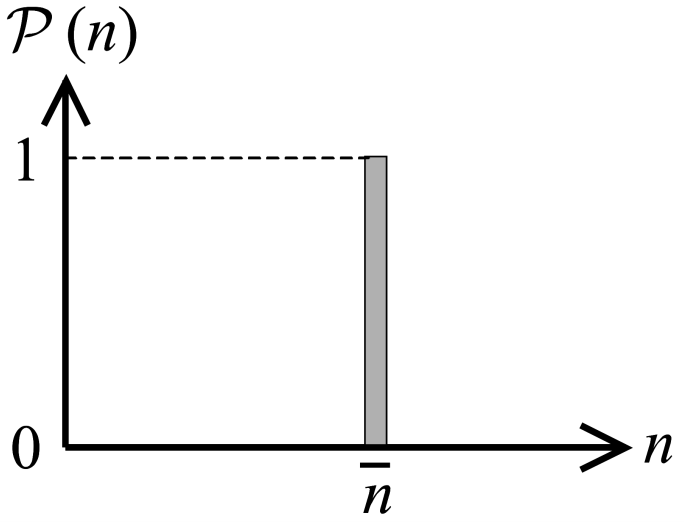
- **Variance**

$$\mathrm{var}\,(n) = \langle \Delta\hat{n}^2 \rangle = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2 = 0$$

- **Variance is less than mean**

↑ Sub-poissonian statistics

Photon number distribution



p. 88 Fox, M. Quantum optics – an introduction (2006)

Sub-poissonian statistics: no semiclassical explanation, provides a clear demonstration that it is light itself that is quantized

Photon statistics

Summary

- **Super-poissonian**

$$\mathrm{var}\,(n) > \bar{n}$$

  - e.g. thermal state $\hat{\rho}_{\mathrm{th}}$

- **Poissonian**

$$\mathrm{var}\,(n) = \bar{n}$$

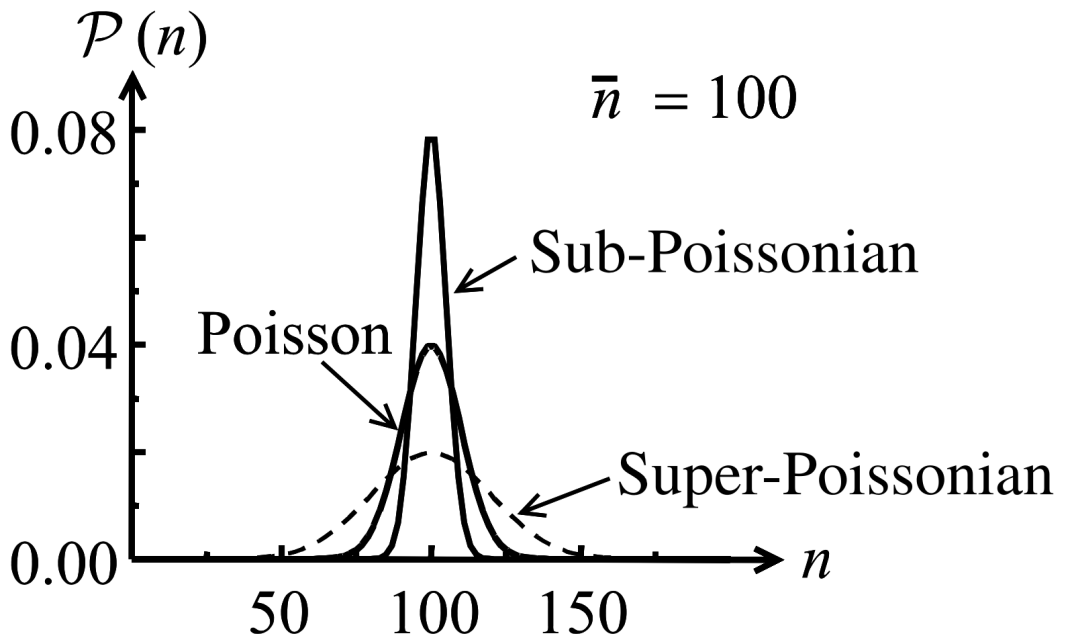
  - e.g. coherent state $|\alpha\rangle$

- **Sub-poissonian**

$$\mathrm{var}\,(n) < \bar{n}$$

  - e.g. photon number state $|n\rangle$

Photon number distribution



p. 82 Fox, M. Quantum optics – an introduction (2006)

Summary

Density operators

To describe mixed ensembles of quantum states, we use density operators:

$$\hat{\rho} = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|$$

p_i: probability of being in state $|\psi_i\rangle$

Thermal states of light

$$\hat{\rho} = \frac{1}{\bar{n}+1} \sum_n \left(\frac{\bar{n}}{\bar{n}+1}\right)^n \left|n\right\rangle \left\langle n\right|$$

Photon statistics

- **Super-poissonian**

$$\mathrm{var}\,(n) > \bar{n}$$

  - e.g. thermal state $\hat{\rho}_{\mathrm{th}}$

- **Poissonian**

$$\mathrm{var}\,(n) = \bar{n}$$

  - e.g. coherent state $\left|\alpha\right\rangle$

- **Sub-poissonian**

$$\mathrm{var}\,(n) < \bar{n}$$

  - e.g. photon number state $\left|n\right\rangle$

- (Photon statistics) Sec. 4.3.3, page 108 of Bachor, Hans-A. / Ralph, Timothy C. A guide to experiments in quantum optics (2019)
- (Photon statistics) Sec 5.1-5.6 of Fox, M. Quantum optics – an introduction (2006)
- (Thermal states) Sec. 3.2.3 page 45 of Leonhardt, U. Essential Quantum Optics (2010)